DEPARTMENT OF COMPUTER ENGINEERING
Course: CPE324 Database Systems
Deadline: October 28, 2026, 17:00

Project Phase 1: E-Commerce Relational Database Design & 3NF

1. Assignment Description & Objective

Design a comprehensive relational schema for a modern multi-vendor e-commerce platform.

Requirements: 1) Construct an Entity-Relationship (ER) diagram with at least 8 entities (Users, Products, Orders, OrderItems, Payments, Categories, Reviews, Shipments).

2) Perform rigorous Database Normalization from 1NF to 3NF, explicitly demonstrating functional dependency closures.

3) Write standard SQL DDL creation scripts with primary and foreign key constraints, indexes, and test data insertion queries.

2. Grading & Evaluation Criteria

• ER Diagram completeness and business logic coverage (35%)

• Mathematical rigor in 3NF Normalization proofs (35%)

• Valid SQL DDL and query performance execution (30%)

Note: Please deconstruct this assignment into daily actionable tasks using TaskFlow AI to avoid procrastination and burnout.